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AN EXACT COUPLING THRESHOLD FOR THE SINGLE-SCALE CURVATURE TEST ON ONE SU(2) LOOP

Main result: for the one-loop Wilson weight $\Omega_g = \exp(\mathrm{Tr}\, U/(2g^2))$ on SU(2), the initial curvature tensor of the multiscale criterion is $\mathrm{Ric} + 2\nabla^2 V_0 = (1 + \mathrm{Tr}\, U/(2g^2))$metric, with optimal lower bound $\kappa_0(g) = 1 - 1/g^2$. The single-scale test therefore fails exactly for $g \le 1$. A Poincaré inequality nevertheless holds for every $g > 0$ with constant $(2/3)e^{2/g^2}$, by direct comparison with Haar measure. Failure of the curvature test shows that this test needs information from positive scales; it does not show that a multiscale argument is needed to prove the inequality.

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Abstract

We examine two claims about the horizontal curvature criterion of [1] in one exactly computable model: one SU(2) loop with the Wilson weight $\Omega_g = \exp(x/g^2)$, $x = \frac{1}{2}\mathrm{Tr}\, U$, and the conjugation gauge action. The first claim is that the single-scale test fails below an explicit coupling. We prove it in sharp form: $\nabla^2 x = -(x/2)\cdot\mathrm{metric}$, so the initial tensor of the criterion is $T_0 = (1 + x/g^2)\cdot\mathrm{metric}$ on all tangent vectors, its optimal horizontal lower bound is $\kappa_0(g) = 1 - 1/g^2$, and the bound is strictly positive exactly when $g > 1$. The adverse configuration is $U = -\mathbb{1}$, where every direction is horizontal and $\log \Omega_g$ has Hessian $1/(2g^2)$. For the classical Bakry–Émery tensor, which carries $\nabla^2 V_0$ instead of $2\nabla^2 V_0$, the threshold is $g = 1/\sqrt{2}$. The second claim is that the failure makes a multiscale step necessary. We show that this holds only in a narrow sense. For $g \le 1$ every admissible curvature profile in the criterion of [1] is non-positive at the initial scale, so that criterion must use positive scales; and it does succeed, with a finite constant, for every $g > 0$. But a Poincaré inequality with constant $(2/3)e^{2/g^2}$ follows for every $g > 0$ from a direct comparison with Haar measure that uses no curvature and no renormalization. Necessity for one sufficient criterion is not necessity for the inequality. We add a one-sided extension to finite lattices: at the configuration with every plaquette equal to $-\mathbb{1}$ the same computation shows that the test fails whenever g^2 is at most the largest eigenvalue of the plaquette–edge incidence form, which lies between 4 and $8(d-1)$ on cubical lattices. The results are elementary. No estimate uniform in the lattice and no statement about a continuum limit is proved.

Origins

This white paper develops the exercise “An exact coupling threshold on one SU(2) loop”. The exercise was built to test a statement proposed earlier in this programme: “the single-scale criterion must fail below an explicit coupling, because on a compact group the Hessian of $\log \Omega$ has positive directions of size about $1/g^2$; this proves the multiscale step is necessary.” The exercise separates that statement into a part that is true in a precise form and a part that is not.

Here	Exercise	Change
Definitions 1–5	Setting	Collected and numbered
Lemma 1: Hessian of x	Part 2, identity $\nabla^2 x = -(x/2)g$	Proved
Lemma 2: adverse configuration	Part 1	Proved
Theorem 3: exact threshold	Part 2	Proved; scalar form of T_0 on all vectors added
Corollary 4: classical tensor	"The factor of two matters"	Threshold $g = 1/\sqrt{2}$ made explicit
Proposition 5: direct comparison	Part 3	Proved; sharp Haar constant 2/3 added
Proposition 6: the criterion succeeds	Closing paragraph	New here: finiteness is automatic on one lattice
Corollary 7: two meanings of "necessary"	Closing paragraph	Stated as a result
Proposition 8: finite lattices	—	New here; one-sided
Remark 9: numerical comparison	—	New here; numerical

0 Definitions

Definition 1 (group and metric). $G = SU(2)$ with the bi-invariant metric induced by $\langle X, Y\rangle = -\mathrm{Tr}(XY)$ on $\mathfrak{su}(2)$. We write g_{met} for the metric tensor, to keep g for the coupling. Then $\mathrm{Ric} = g_{\mathrm{met}}$ [1, Lemma 1], and dU is normalized Haar measure.

Definition 2 (gauge action and horizontal vectors). $SU(2)$ acts on G by conjugation. A tangent vector at U is horizontal if it is orthogonal to the conjugacy class through U. At $U = \pm\mathbb{1}$ the class is a point and every tangent vector is horizontal.

Definition 3 (Wilson weight). $x(U) = \tfrac{1}{2}\mathrm{Tr}\, U \in [-1, 1]$. For a coupling $g > 0$,

$$\Omega_g(U) = \exp(x(U)/g^2), \quad Z_g = \int \Omega_g dU, \quad d\mu_g = Z_g^{-1}\Omega_g dU.$$

In the notation of [1] the action is $S = -\log \Omega_g = -x/g^2$, and $V_0 = S$. The weight and its normalization are part of the model: with the Wilson parameter β of [1, Remark 9], $S = -\beta\, \mathrm{Tr}\, U$, the relation is $\beta = 1/(2g^2)$.

Definition 4 (initial curvature tensors). The initial tensor of the multiscale criterion of [1] is $T_0 = \mathrm{Ric} + 2\nabla^2 V_0 = \mathrm{Ric} - 2\nabla^2 \log \Omega_g$. The classical Bakry–Émery tensor of μ_g is $B_0 = \mathrm{Ric} + \nabla^2 V_0$. The optimal horizontal lower bounds are

$$\kappa_0(g) = \inf\{T_0(h, h) : h \text{ horizontal}, |h| = 1\}, \quad \kappa_{\mathrm{BE}}(g) = \inf\{B_0(h, h) : h \text{ horizontal}, |h| = 1\}.$$

Definition 5 (single-scale test). The single-scale test for T_0 succeeds if $\kappa_0(g) > 0$ and fails otherwise; similarly for B_0. The multiscale criterion is Theorem 5 of [1], an adaptation of the method of [2]: if $(\mathrm{Ric} + 2\nabla^2 V_t)(h, h) \geq \kappa(t)|h|^2$ for horizontal h and all $t \geq 0$, with $V_t = -\log e^{t\Delta}\Omega_g$, then $\mathrm{Var}_{\mu_g}(f) \leq C_\kappa \int |\nabla f|^2 d\mu_g$ for gauge-invariant f, where $C_\kappa = 2\int_0^\infty \exp(-2\int_0^t \kappa) dt$.

1 The adverse configuration

Lemma 1 (Hessian of x). On $SU(2)$, $\nabla^2 x = -(x/2)\, g_{\mathrm{met}}$. Consequently $\Delta x = -(3/2)x$.

Proof. Every $X \in \mathfrak{su}(2)$ is traceless and anti-Hermitian, so by the Cayley–Hamilton theorem $X^2 = \tfrac{1}{2}\mathrm{Tr}(X^2)\,\mathbb{1} = -\tfrac{1}{2}|X|^2\,\mathbb{1}$. The geodesic through U with initial velocity UX is $s \mapsto U e^{sX}$, and

$$(d^2/ds^2)\, x(U e^{sX}) = \tfrac{1}{2}\mathrm{Tr}(U e^{sX} X^2) = -\tfrac{1}{2}|X|^2\, x(U e^{sX}).$$

At $s = 0$ this is $\nabla^2 x(UX, UX) = -(x(U)/2)|X|^2$. A symmetric bilinear form is determined by its quadratic form, so $\nabla^2 x = -(x/2) g_{\mathrm{met}}$. Taking the trace over the three dimensions gives $\Delta x = -(3/2)x$. ∎

Lemma 2 (adverse configuration). Let $U_0 = -\mathbb{1}$ and let $X \in \mathfrak{su}(2)$ with $-\mathrm{Tr}(X^2) = 1$. The unit-speed geodesic $U(s) = -\exp(sX)$ satisfies

$$x(U(s)) = -\cos(s/\sqrt{2}), \qquad (d^2/ds^2) \log \Omega_g(U(s)) \text{ at } s = 0 \text{ equals } 1/(2g^2).$$

Every such direction is horizontal. Hence $\log \Omega_g$ has a positive Hessian of size $1/(2g^2)$ in horizontal directions.

Proof. From $X^2 = -\tfrac{1}{2}\mathbb{1}$, $\exp(sX) = \cos(s/\sqrt{2})\,\mathbb{1} + \sqrt{2}\sin(s/\sqrt{2})X$, and $\mathrm{Tr}\, X = 0$ gives $x(U(s)) = -\cos(s/\sqrt{2})$. Then $\log \Omega_g(U(s)) = -\cos(s/\sqrt{2})/g^2$, whose second derivative at 0 is $1/(2g^2)$. Conjugation fixes $-\mathbb{1}$, so its orbit is a point and every tangent vector is horizontal. ∎

Remark (what compactness does and does not give). Compactness alone does not force a positive Hessian of order g^{-2}: for the constant weight the Hessian of $\log \Omega$ vanishes identically. The size $1/(2g^2)$ is a property of the Wilson weight and of its normalization, and the location $-\mathbb{1}$ is the point where the weight is smallest.

2 The exact threshold

Theorem 3 (exact threshold for the single-scale test). For every $g > 0$:

(i) $T_0 = (1 + x/g^2)\, g_{\mathrm{met}}$ on all tangent vectors.

(ii) $\kappa_0(g) = 1 - 1/g^2$, and the infimum is attained only at $U = -\mathbb{1}$.

(iii) The single-scale test for T_0 succeeds if and only if $g > 1$. At $g = 1$ the tensor T_0 is non-negative and vanishes at $-\mathbb{1}$. For $0 < g < 1$ it is negative definite on the open set $\{x < -g^2\}$.

(iv) If κ satisfies hypothesis (H) of [1] for the action $-\log \Omega_g$, then $\kappa(0) \le 1 - 1/g^2$.

Proof. (i) By Lemma 1, $\nabla^2 \log \Omega_g = g^{-2}\nabla^2 x = -(x/(2g^2))g_{\mathrm{met}}$, and $\mathrm{Ric} = g_{\mathrm{met}}$, so $T_0 = \mathrm{Ric} - 2\nabla^2 \log \Omega_g = (1 + x/g^2)g_{\mathrm{met}}$. (ii) At every U there is a nonzero horizontal vector: the radial direction at regular points and every direction at $\pm\mathbb{1}$. On a unit horizontal vector at U, T_0 takes the value $1 + x(U)/g^2$, and the infimum of this over G is $1 - 1/g^2$, attained exactly where $x = -1$, that is at $U = -\mathbb{1}$. (iii) $1 - 1/g^2 > 0$ if and only if $g > 1$, and $1 +$

$x/g^2 < 0$ if and only if $x < -g^2$, a nonempty open set when $g < 1$. (iv) Hypothesis (H) at $t = 0$ is $T_0(h, h) \geq \kappa(0)|h|^2$ for horizontal h; evaluate it at $-\mathbb{1}$. ∎

Corollary 4 (the classical tensor; the factor two). $B_0 = (1 + x/(2g^2))\, g_{\mathrm{met}}$ and $\kappa_{\mathrm{BE}}(g) = 1 - 1/(2g^2)$. The classical single-scale test succeeds if and only if $g > 1/\sqrt{2}$, and then $\mathrm{Var}_{\mu_g}(f) \leq (1 - 1/(2g^2))^{-1} \int |\nabla f|^2 d\mu_g$ for every smooth f.

Proof. As in Theorem 3 with $\nabla^2 V_0$ in place of $2\nabla^2 V_0$. The inequality is the Bakry–Émery theorem [3, 4] for the generator $\Delta - \langle \nabla V_0, \nabla \cdot \rangle$ of the Langevin dynamics of μ_g; since the bound holds on all tangent vectors, it applies to all f. ∎

The two thresholds are different. For $1/\sqrt{2} < g \leq 1$ the classical test succeeds while the initial tensor of the multiscale criterion is not positive. The threshold $g = 1$ is a statement about T_0, the tensor of [1], and not about every single-scale curvature test.

Remark (the horizontal restriction gives nothing here). T_0 and B_0 are multiples of the metric. Their lower bounds on horizontal vectors and on all vectors coincide, and the adverse point $-\mathbb{1}$ is one where all directions are horizontal. In this model the restriction to horizontal directions, which is the point of the criterion of [1], does not change the initial threshold.

3 What the failure does and does not imply

Proposition 5 (direct comparison). For every $g > 0$ and every smooth f on $SU(2)$, gauge invariant or not,

$$\mathrm{Var}_{\mu_g}(f) \le (2/3)\, e^{2/g^2} \int |\nabla f|^2 \, d\mu_g \le e^{2/g^2} \int |\nabla f|^2 \, d\mu_g .$$

Proof. Since $-1 \le x \le 1$, $e^{-1/g^2} \le \Omega_g \le e^{1/g^2}$ and $\max \Omega_g / \min \Omega_g = e^{2/g^2}$. Let $c = \int f \, dU$. By the variational identity $\mathrm{Var}_\mu(f) = \inf_a \int (f-a)^2 d\mu$,

$$\mathrm{Var}_{\mu_g}(f) \le \int (f-c)^2 d\mu_g \le (\max \Omega_g / Z_g) \int (f-c)^2 dU \le (\max \Omega_g / Z_g)\, C_H \int |\nabla f|^2 dU \le (\max \Omega_g / \min \Omega_g)\, C_H \int |\nabla f|^2 d\mu_g ,$$

where C_H is the Poincaré constant of Haar measure. The smallest positive eigenvalue of $-\Delta$ on $SU(2)$ is $3/2$ [1, Corollary 6], so $C_H = 2/3$; the Bakry–Émery bound from $\mathrm{Ric} = g_{\mathrm{met}}$ gives the weaker value $C_H \le 1$, which is the one used in the exercise. This is the perturbation argument of Holley and Stroock [5]. ∎

The argument uses no curvature of the weight and no renormalization. It gives a finite constant at every fixed $g > 0$, including the whole range $g \le 1$ where $\kappa_0(g) \le 0$. Its constant deteriorates as $g \downarrow 0$.

Proposition 6 (the multiscale criterion succeeds for every g). For every $g > 0$ the optimal function $\kappa_g(t)$ in hypothesis (H) of [1] is continuous on $[0, \infty)$, $\kappa_g(0) = 1 - 1/g^2$, $\kappa_g(t) \to 1$ as $t \to \infty$, and $C_{\kappa_g} < \infty$.

Proof. $z_t = e^{t\Delta}\Omega_g$ is smooth and strictly positive on $[0, \infty) \times G$, and converges to the constant Z_g in C^2 as $t \to \infty$, because every non-constant eigenfunction of Δ decays exponentially. Hence $V_t = -\log z_t$ is smooth, $(t, U) \mapsto \nabla^2 V_t(U)$ is continuous, and $\nabla^2 V_t \to 0$ uniformly. The tensors $\mathrm{Ric} + 2\nabla^2 V_t$ therefore depend continuously on t, uniformly over unit tangent vectors, so the infimum over the fixed set of unit horizontal vectors is continuous in t and tends to 1. Its value at $t = 0$ is given by Theorem 3. A continuous function with limit 1 is locally integrable and exceeds ½ beyond some T, so $\int_0^\infty \exp(-2\int_0^t \kappa_g) dt < \infty$. ∎

The proof uses nothing specific to the Wilson weight and gives no control of the size of C_{κ_g}. On one finite lattice, finiteness of the multiscale constant is automatic; what requires an estimate on V_t for $t > 0$ is its size, and above all its dependence on the lattice. Numerical values are in Remark 9.

Corollary 7 (two meanings of "necessary"). Let $0 < g \le 1$.

(i) Relative necessity, true. Every κ admissible in the criterion of [1] has $\kappa(0) \le 0$. Any finite constant obtained from that criterion must therefore use $\kappa(t) > 0$ on a set of scales $t > 0$ of positive measure, that is, information beyond the initial curvature.

(ii) Absolute necessity, false. A Poincaré inequality for μ_g holds with the explicit constant $(2/3)e^{2/g^2}$, proved without any multiscale analysis.

Proof. (i) Theorem 3(iv) gives $\kappa(0) \le 1 - 1/g^2 \le 0$. If $\kappa \le 0$ almost everywhere then $\int_0^t \kappa \le 0$ for all t and $C_\kappa = \infty$. (ii) Proposition 5. ∎

The failure of one sufficient condition leaves every other route to the same inequality open. The statement "the multiscale step is necessary" is correct when it is read as (i) and incorrect when it is read as a statement about the inequality.

4 Finite lattices and numerical comparison

The computation at $-\mathbb{1}$ extends to several plaquettes. Let (V, E, P) be a finite oriented 2-complex in which the boundary of each plaquette p is a closed path of edges with signs $o_{pe} \in \{0, \pm 1\}$. On $M = SU(2)^E$, with the product metric, let U_p be the ordered holonomy around p and $\Omega_g = \exp(g^{-2}\Sigma_p x(U_p))$. Let d_1 be the $P \times E$ matrix (o_{pe}) and λ_{max} the largest eigenvalue of $d_1^T d_1$.

Proposition 8 (one-sided threshold on a finite lattice). Assume (Z): there is $\eta \in \{0, 1\}^E$ such that $\Sigma_e |o_{pe}| \eta_e$ is odd for every p. Then

$$\kappa_0(g) \le 1 - \lambda_{max}/g^2,$$

so the single-scale test for T_0 fails whenever $g^2 \le \lambda_{max}$. On a cubical complex of dimension d, $4 \le \lambda_{max} \le 8(d-1)$. Hypothesis (Z) holds for open cubical boxes in every dimension.

Proof. Let $U^0_e = (-1)^{\eta_e}\mathbb{1}$. By (Z) every plaquette holonomy of U^0 equals $-\mathbb{1}$. For $A = (A_e) \in \mathfrak{su}(2)^E$ the curve $s \mapsto (e^{sA_e}U^0_e)$ is a geodesic of M. Because the U^0_e are central, the holonomy around p along this curve is $-\Pi e^{\pm sA_e} = -\exp(sY_p + s^2R_p(s))$ with $Y_p = \Sigma_e o_{pe}A_e = (d_1A)_p$ and R_p smooth, by the Baker–Campbell–Hausdorff formula. For $W \in \mathfrak{su}(2)$, $\tfrac{1}{2}\mathrm{Tr}\exp(W) = \cos(|W|/\sqrt{2})$, and $|sY_p + s^2R_p|^2 = s^2|Y_p|^2 + O(s^3)$, so $x(U_p(s)) = -1 + \tfrac{1}{4}s^2|Y_p|^2 + O(s^3)$. Hence

$$\nabla^2 \log \Omega_g(A, A) = (1/(2g^2))\,\Sigma_p |(d_1A)_p|^2, \quad T_0(A, A) = |A|^2 - g^{-2}\,\Sigma_p |(d_1A)_p|^2 \text{ at } U^0.$$

The tangent space to the gauge orbit at U^0 is the image of d_0, $(d_0X)_e = X_{s(e)} - X_{t(e)}$, again because the U^0_e are central. Closed plaquette boundaries give $d_1d_0 = 0$, so every eigenvector of $d_1^T d_1$ with nonzero eigenvalue lies in the image of d_1^T, which is orthogonal to the image of d_0: it is horizontal. Let a be a unit eigenvector for λ_{max} and $X_0 \in \mathfrak{su}(2)$ a unit vector; then $A = a \otimes X_0$ is a unit horizontal vector with $T_0(A, A) = 1 - \lambda_{max}/g^2$.

On a cubical complex each plaquette has four distinct edges and each edge lies in at most $2(d-1)$ plaquettes. The diagonal entries of $d_1d_1^T$ equal 4, so $\lambda_{max} \ge 4$. The absolute row sums of $d_1^T d_1$ are at most $4 \cdot 2(d-1)$, so $\lambda_{max} \le 8(d-1)$. For an open box the constant 2-cochain 1 is closed modulo 2, because every 3-cube has six faces, and the box is contractible, so it is a coboundary; this is (Z). ∎

The bound is one-sided: it shows where the test fails, not where it succeeds, because κ_0 is an infimum over all configurations and only one has been examined. For one loop (one edge, $d_1 = (1)$) it reproduces the threshold of Theorem 3. For open two-dimensional boxes of 1×1, 2×1, 2×2, 3×3, 4×4 and 10×10 plaquettes, $\lambda_{max} = 4, 5, 6, 6.83, 7.24$ and 7.84, approaching the bound 8. On a lattice the test thus fails already for $g^2 \le \lambda_{max}$, a range that extends to stronger coupling than on one loop. This is consistent with the fact that single-scale curvature arguments for lattice Yang–Mills are strong-coupling results [6].

Remark 9 (numerical comparison on one loop). The table lists, for gauge-invariant f, the constants given by each method and the numerically computed optimal constant. The multiscale and optimal columns are computed with the code of [1].

g	$\kappa_0(g)$	Classical single-scale	Direct comparison $(2/3)e^{2/g^2}$	Multiscale C_κ	Optimal (numerical)
2	0.75	1.143	1.099	1.164	0.664
$\sqrt{2}$	0.50	1.333	1.812	1.380	0.656
1	0.00	2.000	4.926	2.078	0.625
$1/\sqrt{2}$	−1.00	fails	36.4	6.55	0.527
1/2	−3.00	fails	1987	222	0.328

Three observations. Where the initial tensor is non-positive ($g \le 1$), both the direct comparison and the multiscale criterion give finite constants, and the multiscale one is the smaller of the two in the tabulated range. Neither is close to the optimal constant, which decreases as g decreases while both bounds grow rapidly. And at $g = 1$, where $\kappa_0 = 0$, the classical single-scale test still gives the constant 2. The multiscale and optimal columns are numerical values, not proved bounds.

5 Status

Result	Status	Scope and remaining input
Lemmas 1–2	Proved	SU(2)
Theorem 3: $\kappa_0(g) = 1 - 1/g^2$, test fails iff $g \le 1$	Proved, sharp	One loop, Wilson weight, tensor T_0 of [1]
Corollary 4: classical threshold $g = 1/\sqrt{2}$	Proved, sharp	One loop, tensor B_0
Proposition 5: Poincaré constant $(2/3)e^{2/g^2}$	Proved	All $g > 0$, all f; not sharp
Proposition 6: multiscale constant finite	Proved	No bound on its size
Corollary 7: relative necessity true, absolute necessity false	Proved	One loop
Proposition 8: failure for $g^2 \le \lambda_{max}$	Proved, one-sided	SU(2), complexes with (Z); exact lattice threshold open
Remark 9: comparison table	Numerical	Multiscale and optimal columns
Proposed statement "this proves the multiscale step is necessary"	Not established	True only as Corollary 7(i)
Uniform estimates, continuum limit, mass gap	Not addressed	—

What the exercise establishes. For the Wilson weight on one loop, the initial curvature test of [1] fails at the explicit threshold $g \le 1$. Any argument that obtains a positive constant from that curvature profile must use information beyond its initial value. That the criterion then succeeds is automatic on one lattice (Proposition 6), but its constant is known only numerically. The claim that multiscale analysis is necessary for any Poincaré bound is false in this model (Proposition 5). The results are elementary consequences of the identity $\nabla^2 x = -(x/2)g_{met}$ and of the classical comparison argument; no novelty is claimed for either ingredient.

References

References marked † are cited from memory and should be checked against the source.

1. R. Cooperman and A.E. Petrunina V.V., "A Conditional Multiscale Horizontal Curvature Criterion on a Finite Gauge Lattice," XISOSIS, Vienna, 11 November 2026, doi:10.5281/zenodo.24137128.
2. R. Bauerschmidt, T. Bodineau and B. Dagallier, "Stochastic dynamics and the Polchinski equation: an introduction," arXiv: 2307.07619.
3. D. Bakry and M. Émery, "Diffusions hypercontractives," Séminaire de Probabilités XIX, Lecture Notes in Mathematics 1123 (1985) 177–206.
4. D. Bakry, I. Gentil and M. Ledoux, *Analysis and Geometry of Markov Diffusion Operators*, Springer (2014).
5. † R. Holley and D. Stroock, "Logarithmic Sobolev inequalities and stochastic Ising models," J. Stat. Phys. 46 (1987) 1159–1194.
6. H. Shen, R. Zhu and X. Zhu, "A stochastic analysis approach to lattice Yang–Mills at strong coupling," arXiv:2204.12737.

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